# Piyumal Sandeepa

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PiyumalSandeepa

## Summary

Undergraduate in Computer Engineering with a strong interest in software development, problem-solving, and machine learning. Skilled in full-stack web development and database management, with hands-on experience building practical applications and exploring data-driven solutions. Actively involved in IEEE volunteering and event coordination, which has helped develop strong leadership, communication, and teamwork skills. Highly motivated, adaptable, and committed to continuous learning, with a focus on applying modern technologies to solve real-world problems.

## Projects

**Early Warning System for Dam Break Flood Emergency (Ongoing | 4-Member Team)** Github

- Developing a web and mobile-based early warning system for dam safety monitoring and emergency flood risk management
- Built a full-stack web and mobile application using React (Vite), Spring Boot, and React Native with TypeScript (TSX), implementing RESTful APIs with a MySQL database and tested using Bruno for real-time monitoring and system integration.
- Implemented community feedback, emergency response guidance, and coordination with authorities.

**Drive lock-Smart Anti-Theft Vehicle Security System (Four-member Team Project)** Github

- Developed a web-based admin dashboard for real-time vehicle monitoring and system management.
- Built the frontend using React and executed the backend using Node.js and Express, with Firebase for real-time data synchronization.
- Integrated IoT devices (ESP32, Raspberry Pi, GPS, RFID) for vehicle tracking and authentication
- Supported system features including GPS tracking, face recognition door unlocks, and vehicle status monitoring via the web dashboard

**RecipeHub – DevOps Project (Individual Project)** Github

- Deployed a containerized full-stack web application on AWS EC2 using Docker and Docker Hub.
- Built automated CI/CD pipelines with Jenkins and GitHub to streamline code integration, build, and deployment.
- Developed using React, Node.js, Express, and MySQL with RESTful API architecture

**Online Concert Ticket Booking System (Individual Project)** Github

- Improved a full-stack event ticket booking web application with user authentication, event browsing, and ticket booking features.
- Built a responsive frontend using React, Vite, JavaScript, and CSS, and a RESTful backend using Node.js and Express.js.
- Applied JWT-based authentication, role-based access control, and secure data handling
- Designed and maintained a MySQL database for users, events, and bookings.

**Lost & Find Mobile Application (Two-Member Team Project)**

Github

- Built a mobile application to help users report lost and found items and enable direct communication between item owners and finders.
- Core features including lost/found item posting, image uploads, item status updates, and user profile management
- Built the application using Flutter, with Firebase Firestore for data storage, Firebase Authentication for user access control, and Firebase Storage for image handling.

**Software Testing & Quality Assurance – To-Do Application (Individual Project)**

- Implemented TDD with JUnit and BDD with Cucumber in a Spring Boot To-Do application to ensure high test coverage and requirement validation.
- Automated UI testing using Selenium and integrated test execution into a GitHub Actions CI pipeline.
- Performed load testing with JMeter and improved code quality through SonarQube analysis.

## Skills

- **Programming Languages**: JavaScript, Java, C++, Python
- **Backend**: Node, Express, Spring Boot
- **Frontend**: React, Tailwind CSS, HTML, CSS
- **Database:** MySQL, Firebase, MongoDB
- **Mobile App Development:** Flutter
- **Tools and Technologies**: GitHub, Figma, Design Patterns, Jira, AWS, Jenkins, Docker

## Education

- **University of Ruhuna**, BSc (Hons) Undergraduate in Computer Engineering
- **Sabaragamuwa University of Sri Lanka**, Certificate in Advanced Java Programming
- **R/Gankanda Central Collage** – **Pelmadulla,** School Education

## Work Experience

- **Internship Trainee**, People's Bank | August 2022 – March 2023

## Volunteering

- **Vice Chairman -** IEEE IAS Ruhuna Student Chapter | February 2026 - Present
- **Program Team Lead -** CareerXpo 2.0, IEEE Student Branch, University of Ruhuna |August 2025 – January 2026
- **PR and Finance Lead**- DesignHer 1.0, WIE Affinity Group - IEEE Student Branch, University of Ruhuna | 2025

## References

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